

Матрично-векторное гур.

$$x \in \mathbb{R}^d, \quad X \in \mathbb{R}^{d \times n}$$

Def. $f: U \rightarrow V$, U, V — ЛНП.

Оператор $df(x): U \rightarrow V$ — производная f в м. $x_0 \in U$,
если $f(x_0 + h) = f(x_0) + df(x)[h] + o(h)$.

— Производная по Фреше

Def. U, V — ЛНП, $Jf(x): U \rightarrow V$:

$$Jf(x)[h] = \lim_{t \rightarrow 0} \frac{f(x + th) - f(x)}{t}, \quad \forall h \in U$$

Производная
по Гато

Производная Фреше $\Rightarrow$ Производная по Гато
 $\Leftarrow$

① В U орт. норм базис $\{e_i\}_{i=1}^n$, тогда $df(x)[e_i]$ —
— i -ая $\mathbb{R}^n$

$$\textcircled{2} \quad df(x)[\underline{h}] = \underbrace{\langle g_f(x), h \rangle}_{\text{yaguent}}$$

Записываем Φ -ую $g(x) = df(x)$

$$\underline{d^2 f(x)[h_1, h_2]} = dg(x)[h_2] = d(df(x)[h_1])[h_2].$$

Второй шаг.

$f: \mathbb{R}^d \rightarrow \mathbb{R}$ — глобальная : $d^2 f(x)[h_1, h_2] = \langle \underbrace{G_f(x)}_{\text{Гессианов}}$

Уточнение : $f: \mathbb{R}^{d \times d} \rightarrow \mathbb{R}^{d \times d}$

$$\partial f_{ijke} = \frac{\partial f_{ij}}{\partial x_{ke}}$$

Вектор

$$1) \quad d(x+y) = dx + dy$$

Матрица

$$d(X+Y) = dX + dY$$

$$2) d(\alpha x) = \alpha dx$$

$$3) d(x^T y) = x^T dy + d(x^T) y$$

$$4) d\left(\frac{f(x)}{g(x)}\right) = \frac{df \cdot g - dg \cdot f}{g^2}$$

5)

6)

$$7) d(\langle x, y \rangle) = \langle x, dy \rangle + \langle dx, y \rangle$$

— " —

— " —

x

$f: U \rightarrow V$

$x \in U$

$$d(Ax) = A dx$$

$$d(x^T) = (dx)^T$$

$$\langle A, B \rangle = \text{tr}(A^T B)$$

$$d(\langle x, y \rangle) = \langle dx, y \rangle + \langle x, dy \rangle$$

$$d(xy) = dx \cdot y + x \cdot dy$$

$$\|A\|_F = \sqrt{\langle A, A \rangle}$$

$$\bullet d(x^{-1}) = -x^{-1}(dx)x^{-1}, \quad x - \text{незав.}$$

$$\bullet d(\det x) = \langle \det(x) x^{-T}, dx \rangle \Rightarrow \nabla(\det(x)) = \det(x) x^{-T}$$

① MHK

$$f(x) = \frac{1}{2} \|Ax - b\|_2^2 = \frac{1}{2} \langle Ax, Ax \rangle - \langle Ax, b \rangle + \frac{1}{2} \|b\|_2^2$$

$$\begin{aligned} (Ax)^T Ax &= (Ax)^T b = \\ &= x^T (A^T Ax) = x^T A^T b \\ &= \langle x, \underbrace{A^T Ax}_{\tilde{A}} \rangle = \langle x, \underbrace{A^T b}_{\tilde{b}} \rangle \end{aligned}$$

$$f(x) = \frac{1}{2} \langle Ax, x \rangle + \langle b, x \rangle + c$$

$$df = \frac{1}{2} \langle d(Ax), x \rangle + \frac{1}{2} \langle Ax, dx \rangle + \langle b, dx \rangle =$$

$$= \frac{1}{2} \langle A dx, x \rangle + \frac{1}{2} \langle Ax, dx \rangle + \langle b, dx \rangle \quad \text{①} \quad \langle g_f, dx \rangle$$

$$\langle A dx, x \rangle = (A dx)^T x = dx^T A^T x = \langle dx, A^T x \rangle$$

$$\text{②} \quad \frac{1}{2} \langle A^T x, dx \rangle + \frac{1}{2} \langle Ax, dx \rangle + \langle b, dx \rangle = \langle \frac{1}{2} (A^T + A)x + b, dx \rangle$$

$$g_f(x) = \frac{1}{2} (A^T + A)x + b$$

$$\nabla f(x) =$$

$$\nabla^2 f(x) = \frac{1}{2} (A^T + A)$$

$$\left\{ \begin{array}{l} A\text{-conv: } \nabla f(x) = Ax + b \\ \nabla^2 f(x) = A \end{array} \right.$$

$$\textcircled{2} \quad f: \mathbb{R}^{n \times d} \rightarrow \mathbb{R}$$

$$f(x) = \text{Tr}(A x^T x), \quad A \in \mathbb{R}^{d \times d}$$

$$\begin{aligned} df(x) &= d(\text{Tr}(A x^T x)) = d(\text{Tr}(I_d \cdot A x^T x)) = d(\langle I_d, A x^T x \rangle) = \\ &= \langle I_d, \underline{d(A x^T x)} \rangle = \langle I_d, A d(x^T) x \rangle + \langle I_d, A x^T d(x) \rangle = \end{aligned}$$

$$\langle I_d, A d(x^T) x \rangle = \text{Tr}(A d(x^T) x) = \text{Tr}(x A d(x^T)) = \langle x A, dx \rangle$$

$$\langle I_d, A x^T dx \rangle = \text{Tr}(\underline{A x^T dx}) = \langle x A^T, dx \rangle$$

$$df(x) = \langle x A, dx \rangle + \langle x A^T, dx \rangle = \langle x (A + A^T), dx \rangle$$

$$\nabla f(x) = xA + xA^T.$$

$$\textcircled{3} \quad \sigma(x) = \frac{1}{1 + \exp(-x)}.$$

Пысьмо $f: \mathbb{R}^n \rightarrow \mathbb{R}$:

$$f(x) = \ln(1 + \exp(\langle x, a \rangle))$$

$$\nabla f(x) = ?$$

$$\nabla^2 f(x) = ?$$

$$\begin{aligned} df(x) &= d(\ln(1 + \exp(\langle x, a \rangle))) = \frac{1}{1 + \exp(\langle x, a \rangle)} \cdot d(1 + \exp(\langle x, a \rangle)) = \\ &= \frac{\exp(\langle x, a \rangle)}{1 + \exp(\langle x, a \rangle)} \cdot \langle a, dx \rangle = \end{aligned}$$

$$= \frac{1}{1 + 1/\exp(\langle x, a \rangle)} = \frac{1}{1 + \exp(-\langle x, a \rangle)} \cdot \langle a, dx \rangle = \underbrace{\sigma(\langle x, a \rangle)}_{\substack{\uparrow \\ dx_1}} \cdot \langle a, dx \rangle$$

$$\nabla f(x) = \sigma(\langle x, a \rangle) \cdot a$$

$$d^2 f(x) = d(\sigma(\langle x, a \rangle) \cdot \underline{\langle a, dx_1 \rangle}) \quad \textcircled{=}$$

$$\left(\frac{1}{1 + \exp(-x)} \right)' = \left((1 + \exp(-x))^{-1} \right)' = + \frac{1}{(1 + \exp(-x))^2} \cdot (+\exp(x)) =$$

$$= \frac{\exp(-x)}{(1 + \exp(-x))^2} = \underbrace{\frac{\exp(-x)}{1 + \exp(-x)}}_{1 - \sigma(x)} \cdot \underbrace{\frac{1}{1 + \exp(-x)}}_{\sigma(x)} = (1 - \sigma(x)) \cdot \sigma(x)$$

$$\textcircled{=} \sigma(\langle x, a \rangle) \cdot (1 - \sigma(\langle x, a \rangle)) \cdot \underbrace{d(\langle x, a \rangle)}_{\substack{\text{=} \\ \langle a, dx \rangle}} \cdot \underline{\langle a, dx_1 \rangle} =$$

$$= [\dots] \cdot \langle a \cdot \langle a, dx_1 \rangle, dx \rangle = [\dots] \cdot \langle aa^T dx_1, dx \rangle$$

$$D^2 f(x) = \sigma(\langle x, a \rangle) \cdot (1 - \sigma(\langle x, a \rangle)) \cdot aa^T.$$

$$\mathbb{R}^d \rightarrow \mathbb{R}^d:$$

$$\frac{\text{softmax}(x)}{s(x)} = \left(\frac{\exp(x_1)}{\sum_{i=1}^d \exp(x_i)}, \dots, \frac{\exp(x_d)}{\sum_{i=1}^d \exp(x_i)} \right)^T$$

$$\frac{\partial s_i}{\partial x_j} \begin{cases} \nearrow i=j \\ \searrow i \neq j \end{cases}$$

$$1) i \neq j: \frac{\partial \left(\frac{\exp(x_i)}{\sum_{k=1}^d \exp(x_k)} \right)}{\partial x_j} = \frac{0 \cdot [\dots] - \exp(x_i) \cdot \exp(x_j)}{(\sum \exp(x_k))^2} =$$

$$= - \frac{\exp(x_i)}{\sum \exp(x_k)} \cdot \frac{\exp(x_j)}{\sum \exp(x_k)} = -s_i \cdot s_j$$

$$2) i=j: \frac{\partial \left(\frac{\exp(x_i)}{\sum \exp(x_k)} \right)}{\partial x_i} = \frac{\exp(x_i) \cdot \sum \exp(x_k) - e^{x_i} \cdot e^{x_i}}{(\sum \exp(x_k))^2}$$

$$= \frac{\exp(x_i)}{\sum \exp(x_k)} - \left(\frac{\exp(x_i)}{\sum \exp(x_k)} \right)^2 = s_i - s_i^2 = s_i(1-s_i)$$

$$[\sigma f(x)]_{ij} = \begin{cases} -s_i \cdot s_j \\ s_i(1-s_i) \end{cases}$$